

7.4 More on the Augmented Matrix

In the first section in this chapter we saw that there were some special cases in the solution to systems of two equations. We saw that there didn't have to be a solution at all and that we could in fact have infinitely many solutions. In this section we are going to generalize this out to general systems of equations and we're going to look at how to deal with these cases when using augmented matrices to solve a system.

Let's first give the following fact.

Fact

Given any system of equations there are exactly three possibilities for the solution.

1. There will not be a solution.
2. There will be exactly one solution.
3. There will be infinitely many solutions.

This is exactly what we found the possibilities to be when we were looking at two equations. It just turns out that it doesn't matter how many equations we've got. There are still only these three possibilities.

Now, let's see how we can identify the first and last possibility when we are using the augmented matrix method for solving. In the previous section we stated that we wanted to use the row operations to convert the augmented matrix into the following form,

$$\left[\begin{array}{cc|c} 1 & 0 & h \\ 0 & 1 & k \end{array} \right] \quad \text{or} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \end{array} \right]$$

depending upon the number of equations present in the system. It turns out that we should have added the qualifier, "if possible" to this instruction, because it isn't always possible to do this. In fact, if it isn't possible to put it into one of these forms then we will know that we are in either the first or last possibility for the solution to the system.

Before getting into some examples let's first address how we knew what the solution was based on these forms of the augmented matrix. Let's work with the two equation case.

Since,

$$\left[\begin{array}{cc|c} 1 & 0 & h \\ 0 & 1 & k \end{array} \right]$$

is an augmented matrix we can always convert back to equations. Each row represents an equation and the first column is the coefficient of x in the equation while the second column is the coefficient of the y in the equation. The final column is the constant that will be on the right side of the equation.

So, if we do that for this case we get,

$$\begin{aligned}(1)x + (0)y &= h &\Rightarrow & x = h \\ (0)x + (1)y &= k &\Rightarrow & y = k\end{aligned}$$

and this is exactly what we said the solution was in the previous section.

This idea of turning an augmented matrix back into equations will be important in the following examples.

Speaking of which, let's go ahead and work a couple of examples. We will start out with the two systems of equations that we looked at in the first section that gave the special cases of the solutions.

Example 1

Use augmented matrices to solve each of the following systems.

(a)
$$\begin{aligned}x - y &= 6 \\ -2x + 2y &= 1\end{aligned}$$

(b)
$$\begin{aligned}2x + 5y &= -1 \\ -10x - 25y &= 5\end{aligned}$$

Solution

(a)
$$\begin{aligned}x - y &= 6 \\ -2x + 2y &= 1\end{aligned}$$

Now, we've **already** worked this one out so we know that there is no solution to this system. Knowing that let's see what the augmented matrix method gives us when we try to use it.

We'll start with the augmented matrix.

$$\left[\begin{array}{cc|c} 1 & -1 & 6 \\ -2 & 2 & 1 \end{array} \right]$$

Notice that we've already got a 1 in the upper left corner so we don't need to do anything with that. So, we next need to make the -2 into a 0.

$$\left[\begin{array}{cc|c} 1 & -1 & 6 \\ -2 & 2 & 1 \end{array} \right] \xrightarrow{R_2 + 2R_1 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -1 & 6 \\ 0 & 0 & 13 \end{array} \right]$$

Now, the next step should be to get a 1 in the lower right corner, but there is no way to do that without changing the zero in the lower left corner. That's a problem, because

we must have a zero in that spot as well as a one in the lower right corner. What this tells us is that it isn't possible to put this augmented matrix form.

Now, go back to equations and see what we've got in this case.

$$\begin{aligned}x - y &= 6 \\ 0 &= 13 \quad ???\end{aligned}$$

The first row just converts back into the first equation. The second row however converts back to nonsense. We know this isn't true so that means that there is no solution. Remember, if we reach a point where we have an equation that just doesn't make sense we have no solution.

Note that if we'd gotten

$$\left[\begin{array}{cc|c} 1 & -1 & 6 \\ 0 & 1 & 0 \end{array} \right]$$

we would have been okay since the last row would return the equation $y = 0$ so don't get confused between this case and what we actually got for this system.

(b)
$$\begin{aligned}2x + 5y &= -1 \\ -10x - 25y &= 5\end{aligned}$$

In this case we **know** from the first section that there are infinitely many solutions to this system. Let's see what we get when we use the augmented matrix method for the solution.

Here is the augmented matrix for this system.

$$\left[\begin{array}{cc|c} 2 & 5 & -1 \\ -10 & -25 & 5 \end{array} \right]$$

In this case we'll need to first get a 1 in the upper left corner and there isn't going to be any easy way to do this that will avoid fractions so we'll just divide the first row by 2.

$$\left[\begin{array}{cc|c} 2 & 5 & -1 \\ -10 & -25 & 5 \end{array} \right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{cc|c} 1 & \frac{5}{2} & -\frac{1}{2} \\ -10 & -25 & 5 \end{array} \right]$$

Now, we can get a zero in the lower left corner.

$$\left[\begin{array}{cc|c} 1 & \frac{5}{2} & -\frac{1}{2} \\ -10 & -25 & 5 \end{array} \right] \xrightarrow{R_2 + 10R_1} \left[\begin{array}{cc|c} 1 & \frac{5}{2} & -\frac{1}{2} \\ 0 & 0 & 0 \end{array} \right]$$

Now, as with the first part we are never going to be able to get a 1 in place of the red zero without changing the first zero in that row. However, this isn't the nonsense that

the first part got. Let's convert back to equations.

$$\begin{aligned}x + \frac{5}{2}y &= -\frac{1}{2} \\ 0 &= 0\end{aligned}$$

That last equation is a true equation and so there isn't anything wrong with this. In this case we have infinitely many solutions.

Recall that we still need to do a little work to get the solution. We solve one of the equations for one of the variables. Note however, that if we use the equation from the augmented matrix this is very easy to do.

$$x = -\frac{5}{2}y - \frac{1}{2}$$

We then write the solution as,

$$\begin{aligned}x &= -\frac{5}{2}t - \frac{1}{2} \\ y &= t\end{aligned}\quad \text{where } t \text{ is any real number}$$

We get solutions by picking t and plugging this into the equation for x . Note that this is NOT the same set of equations we got in the first section. That is okay. When there are infinitely many solutions there are more than one way to write the equations that will describe all the solutions.

Let's summarize what we learned in the previous set of examples. First, if we have a row in which all the entries except for the very last one are zeroes and the last entry is NOT zero then we can stop and the system will have no solution.

Next, if we get a row of all zeroes then we will have infinitely many solutions. We will then need to do a little more work to get the solution and the number of equations will determine how much work we need to do.

Now, let's see how some systems with three equations work. The no solution case will be identical, but the infinite solution case will have a little work to do.

Example 2

Solve the following system of equations using augmented matrices.

$$3x - 3y - 6z = -3$$

$$2x - 2y - 4z = 10$$

$$-2x + 3y + z = 7$$

Solution

Here's the augmented matrix for this system.

$$\left[\begin{array}{ccc|c} 3 & -3 & -6 & -3 \\ 2 & -2 & -4 & 10 \\ -2 & 3 & 1 & 7 \end{array} \right]$$

We can get a 1 in the upper left corner by dividing the first row by a 3.

$$\left[\begin{array}{ccc|c} 3 & -3 & -6 & -3 \\ 2 & -2 & -4 & 10 \\ -2 & 3 & 1 & 7 \end{array} \right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 2 & -2 & -4 & 10 \\ -2 & 3 & 1 & 7 \end{array} \right]$$

Next, we'll get the two numbers under this one to be zeroes.

$$\left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 2 & -2 & -4 & 10 \\ -2 & 3 & 1 & 7 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 - 2R_1 \rightarrow R_2 \\ R_3 + 2R_1 \rightarrow R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 0 & 0 & 12 \\ 0 & 1 & -3 & 5 \end{array} \right]$$

And we can stop. The middle row is all zeroes except for the final entry which isn't zero. Note that it doesn't matter what the number is as long as it isn't zero.

Once we reach this type of row we know that the system won't have any solutions and so there isn't any reason to go any farther.

Okay, let's see how we solve a system of three equations with an infinity number of solutions with the augmented matrix method. This example will also illustrate an interesting idea about systems.

Example 3

Solve the following system of equations using augmented matrices.

$$3x - 3y - 6z = -3$$

$$2x - 2y - 4z = -2$$

$$-2x + 3y + z = 7$$

Solution

Notice that this system is almost identical to the system in the previous example. The only difference is the number to the right of the equal sign in the second equation. In this system it is -2 and in the previous example it was 10 . Changing that one number completely changes the type of solution that we're going to get. Often this kind of simple change won't affect the type of solution that we get, but in some rare cases it can.

Since the first two steps of the process are identical to the previous part we won't discuss them. Here they are.

$$\left[\begin{array}{ccc|c} 3 & -3 & -6 & -3 \\ 2 & -2 & -4 & -2 \\ -2 & 3 & 1 & 7 \end{array} \right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 2 & -2 & -4 & -2 \\ -2 & 3 & 1 & 7 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 - 2R_1 \rightarrow R_2 \\ R_3 + 2R_1 \rightarrow R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -3 & 5 \end{array} \right]$$

We've got a row of all zeroes so we instantly know that we've got infinitely many solutions. Unlike the two equation case we aren't going to stop however. It looks like with a couple of row operations we can make the second column look like it is supposed to in the final form so let's do that.

$$\left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -3 & 5 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 1 & -3 & 5 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1 + R_2 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & -5 & 4 \\ 0 & 1 & -3 & 5 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

In this case we were able to make the second column look like it's supposed to and the third column will never look correct. However, it is possible that the situation could be reversed and it would be the third column that we can make look correct and the second wouldn't look correct. Every system is different.

Once we reach this point we go back to equations.

$$x - 5z = 4$$

$$y - 3z = 5$$

Now, both of these equations contain a z and so we'll move that to the other side in each

equation.

$$x = 5z + 4$$

$$y = 3z + 5$$

This means that we get to pick the value of z for free and we'll write the solution as,

$$x = 5t + 4$$

$$y = 3t + 5$$

$$z = t$$

where t is any real number

Since there are an infinite number of ways to choose t there are an infinite number of solutions to this system.